



A dynamic nonlinear model for a wide-load range operation of ultra-supercritical once-through boiler-turbine units



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ABSTRACT

As the penetration of renewable energy generation increases in power grid, ultra-supercritical (USC) once-through boiler-turbine (OTBT) units are required to operate in a wide load range to maintain the grid stability. Coordinated control system (CCS) coordinates boiler and turbine systems to track load demand and ensure safe and economical operation simultaneously. However, few studies focus on modeling the USC-OTBT units for CCS design under the current stringent requirement. This study develops a dynamic nonlinear model of a USC-OTBT unit operating in a wide load range, which is a first step toward designing an effective CCS. This work first analyzes and captures the nonlinear characteristics of the USC-OTBT unit in low load and extends to model the dynamic process of the unit in a wider load range, from low load to full load. Besides, two energy functions are presented by analyzing open-loop responses to improve the model structure and dynamic modeling accuracy. Model structure is derived from mass and energy conservation laws, and model parameters and functions are identified based on running data. Validation results demonstrate the effectiveness of the model for a wide load range operation of the USC-OTBT unit, providing a basis for an effective CCS design.

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1. Introduction

Recently, the use of renewable energy sources (RESs) has been increasing drastically due to their clean environmental merits, making great economic and environmental contributions globally [1–4]. However, the random and intermittent nature of RESs, such as wind and solar power, has seriously affected operational security and stability of power grid, and thus more than 30% of wind power generation capacity is curtailed in some Chinese provinces since 2010 [2]. To absorb more renewable energy generation into power grid, power system requires ultra-supercritical (USC) once-through boiler-turbine (OTBT) units to regulate unit load rapidly in a wide load range due to their high thermal efficiency, large capacity and low pollutant emission. Coordinated control system (CCS) views boiler and turbine systems as a whole to adjust unit load and maintain safe and economical operation simultaneously. Due to the intermittent and random nature of RES, it is challenging to design satisfactory CCS of USC-OTBT units to meet the strict requirements of power grid. As is known, satisfactory control performance

depends on understanding of industrial processes, and a simple and accurate model is conducive to operational analysis and control system design.

For USC-OTBT units, the modeling methods can roughly be divided into two categories: first-principle and data-driven modeling methods. First-principle modeling method adopts mass and energy conservation laws to analyze operational process, and then physical model structure is developed based on certain simplifications and assumptions. Parameters and functions in model structure are identified by combining running data and optimization algorithm. The advantages of this modeling method are that the model established has a simple model structure and proper modeling accuracy, and parameters and functions have relatively explicit physical meanings. Importantly, due to mechanism analysis, the model can reflect essential dynamic characteristics in abnormal conditions, and then reveal valuable information. Therefore, this modeling method is conducive to operational analysis and control system design [5–10]. However, due to the complex nature of industrial processes, it is difficult for first-principle modeling method to describe the operational process accurately based on limited knowledge, and thus the model generally does not have high modeling accuracy.

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Nomenclature			
u_B	Coal flow command (kg/s)	c	Specific heat capacity (kJ/(kg·°C))
N_e	Unit load (MW)	V	Boiler volume (m ³)
k_1	Heat energy absorbed by working fluid from burning 1 kg coal in regular operation (kJ/kg)	m	mass (kg)
r_B	Pulverized-coal flow in furnace (kg/s)	Subscripts	
NaN	Not a number	m	State in separator
c_0	Milling inertia time (s)	st	State at throttle valve
Q_1	Energy absorbed by working fluid (kJ/s)	fw	Feed water
Q_{loss}	Throttle loss (MW)	j	State in metal
l	Ratio of main steam enthalpy to steam enthalpy in separator	r	Set point
D	Steam or water mass flow rate (kg/s)	Abbreviations	
h	Specific enthalpy (kJ/kg)	OTBT	Once-through boiler-turbine
p	Steam pressure (MPa)	CCS	Coordinated control system
R^2	Coefficient of determination	USC	Ultra-supercritical
τ	Delay time of pulverized-coal system (s)	SST	Superheated steam temperature
k_2	Turbine coefficient	RES	Renewable energy source
ρ	Steam density (kg/m ³)	THA	Turbine heat-acceptance
α	Designed ratio of water to coal	MAE	Mean absolute error
u_t	Throttle valve opening ($0 \leq u_t \leq 1$)	MARE	Mean absolute relative error
		RMSE	Root mean square error

Liu, Yan and Zeng [11] adopted the first-principle modeling method to develop a dynamic model of a 1000 MW USC-OTBT unit, where the once-through boiler is treated as a whole to reduce the model complexity. The model captures the essential characteristics, and it can be used for controller design. However, this work ignores the dynamic process of turbine system and lacks dynamic validation in a wide load range, which increases the difficulty in control [8,9]. To improve modeling accuracy and consider the influence of heat storage in metal, Fan, Zhang and Su [12] built a dynamic mathematical model of a USC-OTBT unit under dry operation, where heat energy stored in metal is captured in the model. Besides, several groups of dynamic parameters are identified by combining running data and improved immune genetic algorithm to enhance modeling accuracy. However, the complexity of model structure is increased by adding the function of energy storage in boiler metal and many groups of dynamic parameters, which requires much computational cost and is not conducive to control system design. To build up a control-oriented model with appropriate accuracy, Tian et al. [5] developed a nonlinear control-oriented model for a 1000 MW coal-fired unit based on mechanism analysis and model reduction technique. The model is used for designing sliding mode predictive controller, and simulation results illustrate the effectiveness of the model and control scheme. However, this model structure needs to be further improved to reflect proper characteristics in abnormal conditions. As OTBT units usually operate in low load to absorb RESs into power grid, it is necessary to investigate the operational characteristics of OTBT units at low load condition. Thus, Niu et al. [13] presented a dynamic mathematical model of a 600 MW OTBT unit in low load. Model structure is derived from mass and energy conservation laws, and simulation results show that the model has acceptable accuracy. However, the model has a complex structure, and it is not conducive to controller design. As superheated steam temperature (SST) system interacts with CCS, it is essential to investigate their relationship and improve control performances. Therefore, Fan, Su and Wang [1] developed a dynamic mathematical model for a USC-OTBT unit with SST. This work established a seventh-order dynamic model containing spray water and steam temperature, which can reflect the essential interactions between CCS and SST system. The

complexity of model structure is increased again by considering SST, which increases the difficulty in CCS design.

Different from the first-principle modeling method, data-driven modeling method discards mechanism analysis and focuses on running data. Generally, data-driven model has satisfactory modeling accuracy by comparing with first-principle models, but model performance may deteriorate in abnormal conditions due to lack of *a priori* mechanism analysis. In recent years, this modeling method adopts fuzzy structure [14–16], neural networks [14,17–19], recursive least squares [14] and transfer function matrix [20] to model USC-OTBT units. For the fuzzy and neural networks models, the model structure is complex and unreliable, which is difficult to understand the operational process and increases the difficulty in designing control system. Compared with fuzzy and neural networks models, linear modeling methods, such as recursive least squares [14] and transfer function matrix [20], cannot describe the dynamic nonlinear process of USC-OTBT units in a wide load range accurately, but the model structure is relatively simple and conducive to control system design.

Up to date, no relevant models can describe the dynamic process of USC-OTBT units in a wide load range from low load to full load, which is not conducive to understanding the operational process and designing a satisfactory CCS of USC-OTBT units in order to meet strict requirements from power grid. Due to the lack of mechanism analysis, data-driven models do not have reliable structure to reflect the essential dynamic characteristics in practice, which is critically important to control system design. Therefore, this study adopts the first-principle modeling method to establish such a dynamic nonlinear model for a wide-load range operation of USC-OTBT units. First, this study analyzes the nonlinear characteristics of USC-OTBT units in low load and expands it to describe the dynamic process of USC-OTBT units in a wide load range. Second, to improve model performance, two energy functions are presented by analyzing open-loop responses [11], which yields higher modeling accuracy in normal and abnormal conditions. Third, this model can also provide parameter information to understand the operational process of USC-OTBT units in steady-state conditions, and more importantly, input set points can be generated from the given output set points, which are then treated as feedforward

control signals to improve control performance. Finally, validation results demonstrate the effectiveness of the model, and control performance illustrates that the model can be used for CCS design in a wide load range.

The rest of this paper is organized as follows: Section 2 introduces the study plant, a 1000 MW USC-OTBT coal-fired unit. Section 3 develops a dynamic nonlinear model of the unit, and parameters and functions in model are identified in Section 4. In Section 5, open-loop validation is conducted to demonstrate the effectiveness of model structure, and dynamic validation is performed to illustrate an acceptable modeling accuracy. In addition, input set points are generated from the given output set points, and nonlinear model predictive control (NMPC) is used to test the model and demonstrate its control performance. Section 6 concludes the paper and gives research directions in future.

2. Description of USC-OTBT coal-fired units

This work considers a 1000 MW USC-OTBT coal-fired unit shown in Fig. 1 as a study unit. The unit is an energy conversion device consisting of a USC sliding-pressure once-through boiler and a USC single reheat steam turbine. For working process, raw coal is ground in pulverizing system and then sent to furnace for combustion and working fluid in boiler system absorbs heat energy released from coal combustion to generate qualified steam. Subsequently, the steam expands in turbine to generate mechanical energy, and then mechanical energy is converted to electricity in generator. Table 1 shows the key parameters of the unit.

For USC-OTBT units, the major task of CCS is to regulate unit load and then meet strict requirements from power grid while maintaining the safe and economical operation. In practice, coal flow command controls pulverized coal flow and then provides enough heat energy in furnace [20]. Feedwater flows through boiler to absorb the heat energy released by coal combustion and then generate the qualified steam [20]. To ensure the operational safety, steam enthalpy in separator can reflect the change of ratio of coal to water sensitively [11–13], and it is usually controlled within a certain range in practice to maintain a safe operation of boiler system. Besides, main steam pressure can represent the energy balance between boiler and turbine systems, and it should be regulated at a certain value based on current unit load. Throttle valve opening can change the flowing area of working fluid quickly

Table 1

Key parameters at Turbine Heat-Acceptance (THA) condition.

Item	Value
Nominal power (MW)	1000
Superheated steam flow (t/h)	2732
Superheated steam pressure (MPa)	26.09
Superheated steam temperature (°C)	605
Reheated steam flow (t/h)	2276
Water temperature at inlet of economizer (°C)	291
Pressure at inlet of economizer (MPa)	29.65
Condenser pressure (kPa)	5.2
Boiler volume (m ³)	639

[20], and thus adjust main steam pressure. Similarly, energy in boiler system can be stored or released by regulating the opening to track load command rapidly. Accordingly, CCS model [5,11–19] has three inputs, namely, coal flow command, feedwater flow and throttle valve opening, and three outputs, i.e., main steam pressure, steam enthalpy in separator and unit load. Fig. 2 presents the relationship between model inputs and outputs.

3. Modeling for USC-OTBT coal-fired units

Before modeling the units, there are some assumptions shown as follows:

- (A1) Ignore the time delay (200 ms) [5] of combustion process in furnace.

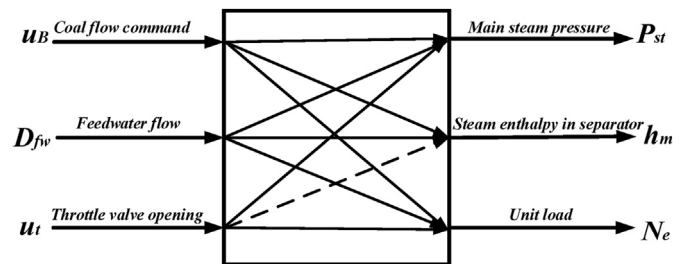


Fig. 2. A structure diagram of CCS model of a USC-OTBT unit (solid line: strong interaction; dashed line: weak interaction).

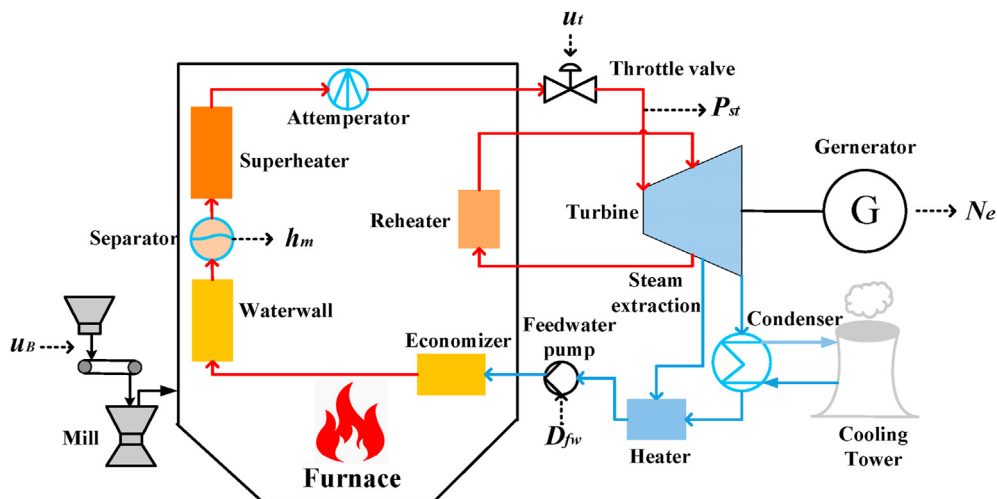


Fig. 1. Schematic diagram of a 1000 MW USC-OTBT coal-fired unit (black solid line: coal flow; red solid line: steam or water flow in boiler system; blue solid line: steam or water flow in turbine system). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

- (A2) There is only a whole-heated tube where working fluid flows through directly under dry operating condition, and it consists of economizer, waterwall, separator and superheater [11,13].
- (A3) Heat energy absorbed by working fluid is approximately proportional to the heat energy released from coal combustion.
- (A4) Heat energy released from coal combustion is distributed in the boiler tube uniformly, and the varying rate of metal temperature in boiler tube is equal to the rate of steam temperature in boiler system [1].
- (A5) Working fluid has uniform properties in the cross section of tube [13].

3.1. Pulverizing system

For simplicity, the dynamic process of pulverizing system can be described as a first-order inertia section with time delay [1,5,11–13]:

$$\frac{dr_B}{dt} = -\frac{1}{c_0}r_B + \frac{e^{-\tau s}}{c_0}u_B, \quad (1)$$

where s is the Laplace operator.

3.2. Boiler system

Due to sensitivity of steam enthalpy in separator, steam states in separator are treated as lumped parameters to model boiler system. According to mass and energy conservation laws, the dynamic process of boiler system can be presented as below:

$$V_m \frac{d\rho_m}{dt} = D_{fw} - D_{st}, \quad (2)$$

$$V_m \frac{d\rho_m h_m}{dt} + c_j m_j \frac{dT_j}{dt} = D_{fw} h_{fw} - D_{st} h_{st} + Q_1, \quad (3)$$

$$Q_1 = k_1 r_B + \frac{\mu}{N_e + \gamma} (D_{fw} - \alpha r_B), \quad (4)$$

where $V_m \frac{d\rho_m}{dt}$ and $V_m \frac{d\rho_m h_m}{dt} + c_j m_j \frac{dT_j}{dt}$ are the mass change (kg/s) of working fluid and energy change (kJ/s) of working fluid and metal tube, respectively.

Remark 1. For USC-OTBT units, coal flow and feedwater flow keep an approximate ratio, namely, 1:7, to provide enough heat energy in furnace and maintain safe operation. However, the ratio varies in practice due to varying coal quality and combustion conditions. In previous works [1,5,11–13], the heat energy absorbed by working fluid is $Q_1 = k_1 r_B$, where k_1 is a function of pulverized-coal flow in furnace, r_B , determined in steady-state conditions. However, this function cannot describe the heat energy absorbed by working fluid accurately since coal flow cannot keep the fixed ratio of feedwater flow. Besides, k_1 is difficult to calculate due to unmeasured state variable, r_B , in practice. In this study, the heat energy absorbed by working fluid is modified as (4) by augmenting with a energy

mismatch function $\mu(D_{fw} - \alpha r_B)/(N_e + \gamma)$, where μ and γ are constants, and reasons for the modification are given as follows:

- (1) k_1 is defined as a function of unit load, N_e , namely, $k_1 = k(N_e)$, and thus can be calculated and used for control system design. The first term in (4), $k_1 r_B$, represents heat energy absorbed by working fluid under the condition that coal flow keeps the fixed ratio of feedwater flow.
- (2) In the second term of (4), $(D_{fw} - \alpha r_B)$ represents the mismatch between feedwater flow and coal flow, where α is the ratio of feedwater flow to coal flow in steady-state condition, and it is a function of unit load, N_e .
- (3) The effect of the mismatch on the change of heat energy is different for different load conditions, and it becomes less severe as load increases. Therefore, $\mu(D_{fw} - \alpha r_B)/(N_e + \gamma)$ represents the deviation of heat energy absorbed by working fluid under the condition that coal flow cannot keep the fixed ratio of feedwater flow.

Thus, the heat energy absorbed by working fluid defined by (4) can describe the change of the heat energy during practical operation and further improve model performance.

According to assumptions, (A4) and (A5), (2) and (3) respectively can be rewritten as

$$V_m \left(\frac{\partial \rho_m}{\partial p_m} \frac{dp_m}{dt} + \frac{\partial \rho_m}{\partial h_m} \frac{dh_m}{dt} \right) = D_{fw} - D_{st}, \quad (5)$$

$$V_m \left\{ h_m \frac{\partial \rho_m}{\partial p_m} \frac{dp_m}{dt} + \left(h_m \frac{\partial \rho_m}{\partial h_m} + \rho_m \right) \frac{dh_m}{dt} \right\} + c_j m_j \left(\frac{\partial T_m}{\partial p_m} \frac{dp_m}{dt} + \frac{\partial T_m}{\partial h_m} \frac{dh_m}{dt} \right) = D_{fw} h_{fw} - D_{st} h_{st} + Q_1, \quad (6)$$

which can be further expressed as

$$b_{11} \frac{dp_m}{dt} + b_{12} \frac{dh_m}{dt} = D_{fw} - D_{st}, \quad (7)$$

$$b_{21} \frac{dp_m}{dt} + b_{22} \frac{dh_m}{dt} = D_{fw} h_{fw} - D_{st} h_{st} + Q_1, \quad (8)$$

$$\begin{aligned} \text{where } b_{11} &= V_m \frac{\partial \rho_m}{\partial p_m}, b_{12} = V_m \frac{\partial \rho_m}{\partial h_m}, b_{21} = V_m h_m \frac{\partial \rho_m}{\partial p_m} + c_j m_j \frac{\partial T_m}{\partial p_m}, b_{22} \\ &= V_m \left(h_m \frac{\partial \rho_m}{\partial h_m} + \rho_m \right) + c_j m_j \frac{\partial T_m}{\partial h_m}, \end{aligned}$$

and, $h_{st} = l h_m$.

Combining (7) and (8), $\frac{dp_m}{dt}$ and $\frac{dh_m}{dt}$ can be calculated as below:

$$c_1 \frac{dp_m}{dt} = (h_{fw} - d_1) D_{fw} + (d_1 - h_{st}) D_{st} + Q_1, \quad (9)$$

$$c_2 \frac{dh_m}{dt} = (h_{fw} - d_2) D_{fw} + (d_2 - h_{st}) D_{st} + Q_1, \quad (10)$$

where

$$\begin{cases} c_1 = b_{21} - \frac{b_{11}b_{22}}{b_{12}} = -\rho_m V_m \frac{\partial \rho_m}{\partial p_m} / \frac{\partial \rho_m}{\partial h_m}, & c_2 = b_{22} - \frac{b_{12}b_{21}}{b_{11}} = \rho_m V_m, \\ d_1 = \frac{b_{22}}{b_{12}} = h_m + \rho_m / \frac{\partial \rho_m}{\partial h_m} + \frac{c_j m_j}{V_m} \frac{\partial T_m / \partial h_m}{\partial \rho_m / \partial h_m}, & d_2 = \frac{b_{21}}{b_{11}} = h_m + \frac{c_j m_j}{V_m} \frac{\partial T_m / \partial p_m}{\partial \rho_m / \partial p_m}. \end{cases} \quad (11)$$

Remark 2. In previous work [1,11–13], dynamic parameters, c_1 , c_2 , d_1 and d_2 , are treated as constants identified by running data and optimization algorithm, and their physical meanings are ignored. To reflect the actual model, c_1 , c_2 , d_1 and d_2 are calculated in this study by using basic information in units and IAPWS-IF97 [21].

Based on previous studies [1,11,12], differential pressure of steam in boiler is expressed as

$$\Delta p = g(p_m), \quad (12)$$

where $g(\cdot)$ is a nonlinear function, and $\Delta p = p_m - p_{st}$ is steam pressure drop in superheater, MPa.

3.3. Turbine system

In turbine system, superheated steam drives turbines to generate mechanical work, and then mechanical work is converted into electrical power in generator. Besides, turbine regenerative system uses steam extracted from turbines to heat the condensate water to improve boiler efficiency.

Superheated steam flow [1,11,12] can be defined as

$$D_{st} = u_t f(p_{st}, h_{st}), \quad (13)$$

where u_t is throttle valve opening and $f(\cdot, \cdot)$ is a nonlinear function.

Dynamic process of turbine [1,5,13] is presented as

$$\frac{dN_e}{dt} = \frac{1}{c_3} k_2 D_{st} (h_{st} - h_{fw}) - \frac{1}{c_3} N_e, \quad (14)$$

where k_2 is a nonlinear function, $k_2 = k(N_e)$, and c_3 is inertia time of turbine system, s.

Compared with previous work [1,5,11–13], throttle loss in turbine system is first considered in this study. Throttle loss (Q_{loss}) is energy loss due to the drop of steam pressure while superheated steam flows through throttle valve in the high-pressure (HP) turbine. The change of the throttle loss, ΔQ_{loss} , has an influence on the dynamic response of the unit load, and needs to be considered to improve model performance. Up to now, there is no relevant work that describes the throttle loss in the CCS model. In practice, ΔQ_{loss} has a close relationship with the change of throttle valve opening, Δu_t , and current steam pressure, p_{st} , and thus a change function of

throttle loss can be defined as $\Delta Q_{loss} = q(\Delta u_t, p_{st}) = \eta \Delta u_t p_{st}$, where Δu_t is the change of varying throttle valve opening, and η is a constant. Therefore, (14) is modified as

$$\frac{dN_e}{dt} = \frac{1}{c_3} \left(k_2 D_{st} (h_{st} - h_{fw}) + \Delta Q_{loss} \right) - \frac{1}{c_3} N_e \quad (15)$$

3.4. Discussion

Previous models [1,5,11–13] are developed based on the first-principle method, and one of major advantages of the first-principle model is to reflect the dynamic characteristics of operational process in abnormal conditions. In engineering practice, transfer functions are usually identified in the abnormal condition that step or ramp disturbance is injected into one input variable of an operational process at steady-state condition to obtain dynamic responses of output variables. Subsequently, parameters of PID controller can be adjusted based on the transfer function of the operational process to design control system. However, previous models do not have good model performances in abnormal conditions, which is not conducive to understanding practical operational process and designing a satisfactory control system. To overcome this problem, we investigated the actual dynamic characteristics [11] of USC-OTBT units in abnormal conditions, and found the important influence of the mismatch between feedwater flow and coal flow and the change of throttle loss on model performance in abnormal conditions. These effects exist in practice indeed but are ignored in previous models [1,5,11–13]. In this work, we consider these factors and first establish two energy functions to describe extra physical information and then improve model construction. Therefore, this model can be used for simulation study to understand the operational process, and then promote control performance to meet strict requirements from power grid.

4. Summary

From the above models for boiler and turbine systems, the model structure of the USC-OTBT unit can be represented in the form of a vector state-space equation as follows:

Table 2
Steady-state data at different loads.

N_e/MW	$u_B/(\text{kg/s})$	$D_{fw}/(\text{kg/s})$	u_t	$D_{st}/(\text{kg/s})$	$h_{fw}/(\text{kJ/kg})$	$h_m/(\text{kJ/kg})$	P_m/MPa	P_{st}/MPa	$T_{st}/^\circ\text{C}$	$h_{st}/(\text{kJ/kg})$
344.0	37.1	306.0	0.6845	305.8	974.1	2768.3	9.955	9.464	475.2	3317.7
389.8	46.7	348.9	0.6860	349.0	1036.9	2808.8	11.324	10.724	473.0	3294.1
433.8	54.4	381.3	0.6861	381.4	1072.3	2806.7	12.480	11.823	487.2	3317.5
498.2	57.1	407.6	0.6766	407.5	1103.9	2827.6	14.092	13.357	537.5	3434.3
600.9	65.5	452.8	0.6733	452.8	1147.3	2880.6	16.462	15.589	600.0	3578.1
878.0	95.6	666.7	0.6695	666.7	1255.7	2791.1	24.043	22.795	602.9	3522.3
951.5	102.7	723.7	0.6644	723.7	1280.8	2790.3	26.148	24.808	602.9	3504.1
1047.3	116.7	803.9	0.6763	803.9	1312.5	2738.7	28.377	26.883	602.8	3484.1

$$\begin{cases} \dot{X} = A(X) + B(X, U), \\ Y = C(X), \end{cases} \quad (16)$$

where U , Y and X are inputs, outputs and state vectors defined by

$$U = [u_1, u_2, u_3]^T = [u_B, D_{fw}, u_t]^T; \quad Y = [y_1, y_2, y_3]^T = [p_{st}, h_m, N_e]^T; \\ X = [x_1, x_2, x_3, x_4]^T = [r_B, p_m, h_m, N_e]^T.$$

The details of the vector state-space model is summarized below:

$$\begin{aligned} \dot{x}_1 &= -\frac{1}{c_0}x_1 + \frac{e^{-\tau s}}{c_0}u_1 \\ \dot{x}_2 &= \frac{h_{fw} - d_1}{c_1}u_2 + \frac{d_1 - h_{st}}{c_1}D_{st} + \frac{Q_1}{c_1} \\ \dot{x}_3 &= \frac{h_{fw} - d_2}{c_2}u_2 + \frac{d_2 - h_{st}}{c_2}D_{st} + \frac{Q_1}{c_2} \\ \dot{x}_4 &= \frac{x_4}{c_3} + \frac{k_2(h_{st} - h_{fw})D_{st} + \Delta Q_{loss}}{c_3} \\ y_1 &= x_2 - g(x_2) \\ y_2 &= x_3 \\ y_3 &= x_4 \end{aligned}$$

where, $D_{st} = u_3 f(p_{st}, h_{st})$, $p_{st} = x_2 - g(x_2)$, $h_{st} = lh_m$, $Q_1 = k_1 r_B + \frac{\mu}{N_e + \gamma}(D_{fw} - \alpha r_B)$, $\Delta Q_{loss} = \eta \Delta u_3 p_{st}$.

5. Model parameter identification

In the model (16), there are four static parameters, h_{fw} , l , k_1 and k_2 ; three dynamic parameters, τ , c_0 and c_3 ; two nonlinear functions, $f(\cdot, \cdot)$ and $g(\cdot)$; and two energy functions, change of throttle loss and energy mismatch functions that need to be identified. Static parameters are the parameters that can be identified in steady-state conditions, while dynamic parameters are the parameters that can only be identified under dynamic conditions. As for data used for identifying parameters, running data for energy functions are collected from previous work [11], and running data in a wide load range are from the study unit, a 1000 MW coal-fired unit, presented in Data Availability.

5.1. Static parameter identification

To obtain static parameters in the model, steady-state data at different loads are first obtained as presented in Table 2.

By setting the derivatives equal to zero in the model (16), the following equations for steady-state conditions can be found to

Table 3
Static parameters at different loads.

Load/(MW)	Static parameters		
	k_1	k_2	l
344.0	19330.0	0.4800	1.19846
389.8	16863.7	0.4948	1.17278
433.8	15737.0	0.5066	1.81993
498.2	16635.2	0.5246	1.21456
600.9	16804.1	0.5459	1.24214
878.0	15806.9	0.5810	1.26198
951.5	15667.0	0.5914	1.25581
1047.3	14963.4	0.5997	1.27239

Table 4
Ratio of water to coal at different loads.

N_e /(MW)	344.0	389.8	433.8	498.2	600.9	878.0	951.5	1047.3
D_{fw}/u_B	8.248	7.471	7.009	7.138	6.913	6.974	7.047	6.889
$T_{st}/^\circ\text{C}$	475.2	473.0	487.2	537.5	600.0	602.9	602.9	602.8

calculate the static parameters:

$$0 = u_B^* - r_B^*, \quad (17)$$

$$0 = D_{fw}^* - D_{st}^*, \quad (18)$$

$$0 = D_{fw}^* h_{fw}^* - D_{st}^* h_{st}^* + k_1 r_B^*, \quad (19)$$

$$0 = k_2 D_{fw}^* (h_{st}^* - h_{fw}^*) - N_e^*, \quad (20)$$

where superscript * means variables lie in steady-state conditions.

Solving (17)–(20), static parameters can be obtained in an explicit form as below:

$$l = \frac{h_{st}^*}{h_m^*}, \quad (21)$$

$$k_1 = \frac{D_{st}^* h_{st}^* - D_{fw}^* h_{fw}^*}{u_B^*}, \quad (22)$$

$$k_2 = \frac{N_e^*}{D_{st}^* (h_{st}^* - h_{fw}^*)}. \quad (23)$$

By combining (21)–(23) and steady-state data in Table 2, static parameters can be calculated as presented in Table 3.

As shown in Table 3, static parameters change with unit load, and thus the parameters can be identified as nonlinear functions of unit load (N_e). It should be noted that four static parameters are nonlinear functions of unit load in this study, and the model structure can provide more information used for controller design presented in the next section.

As presented in Table 4, the ratio of water to coal, D_{fw}/u_B , increases rapidly as the load decreases from 433.8 MW to 344.0 MW,

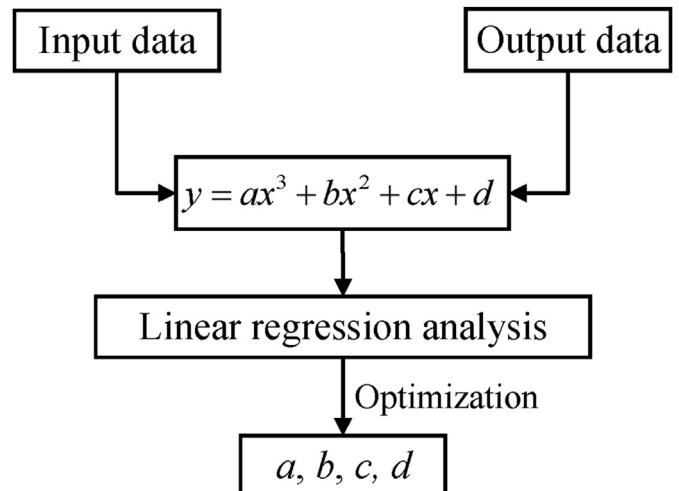


Fig. 3. Procedure of regression parameter estimation.

and superheated steam temperature, $T_{st}/^{\circ}\text{C}$, cannot reach the designed rated value in low load. Therefore, a nonlinearity exists in a low load range roughly from 35% to 50% of the nominal load. According to operational experience, combustion condition deteriorates in low load; furnace temperature decreases and energy loss increases. Considering operational safety, it is necessary to provide more feedwater flow in order to maintain steam flowing in boiler tube and avoid metal overheat. Meanwhile, superheated steam temperature decreases in low load, and boiler and turbine efficiency decreases simultaneously. This means that coal consumption rate increases in low load, which can affect functional relationship of variables existing in the model.

To improve modeling accuracy, a cubical polynomial of the unit load is used to describe the nonlinearity of each of the static parameters, h_{fw} , k_1 , k_2 and l , as below:

$$\begin{cases} h_{fw} = -5.4 \times 10^{-9}N_e^4 + 1.613 \times 10^{-5}N_e^3 - 1.76 \times 10^{-2}N_e^2 + 8.655N_e - 499.83, \\ k_1 = 4.247 \times 10^{-7}N_e^4 - 1.248 \times 10^{-3}N_e^3 + 1.315N_e^2 - 587.15N_e + 110301.62, \\ k_2 = 3.8 \times 10^{-17}N_e^4 + 1.2646 \times 10^{-13}N_e^3 - 5.48 \times 10^{-10}N_e^2 + 6.7 \times 10^{-7}N_e + 3.085 \times 10^{-4}, \\ l = 8.375 \times 10^{-12}N_e^4 - 2.365 \times 10^{-8}N_e^3 + 2.37 \times 10^{-5}N_e^2 - 9.747 \times 10^{-3}N_e + 2.588. \end{cases}$$

$$y = ax^3 + bx^2 + cx + d \quad (24)$$

where a , b , c and d are regression parameters, x is the unit load, and y represents static parameters, h_{fw} , k_1 , k_2 or l .

The regression parameters can be estimated based on linear regression analysis with data in Tables 2 and 3 Fig. 3 presents the procedure of regression parameter estimation.

For static parameters, h_{fw} , l , k_1 and k_2 , input data in Fig. 3 are a set of unit load, N_e , in the range from 344.0 MW to 1047.3 MW in Table 2, and similarly output data, h_{fw} , l , k_1 and k_2 , are given in Tables 2 and 3 For the four static parameters, the results of regression parameter estimation are presented as below:

$$h_{fw} = 9.778 \times 10^{-7}N_e^3 - 2.4 \times 10^{-3}N_e^2 + 2.27N_e + 449.3, \quad (25)$$

$$k_1 = -5.55 \times 10^{-5}N_e^3 + 0.12N_e^2 - 84.87N_e + 35634.5, \quad (26)$$

$$k_2 = 2.341 \times 10^{-13}N_e^3 - 6.56 \times 10^{-10}N_e^2 + 7.15 \times 10^{-7}N_e + 3.02 \times 10^{-4}, \quad (27)$$

$$l = -1.383 \times 10^{-10}N_e^3 + 1.314 \times 10^{-7}N_e^2 + 1.578 \times 10^{-4}N_e + 1.116. \quad (28)$$

The nonlinearities of parameters are all assumed to be the third-order polynomials in (24). However, it is necessary to analyze the influence of the order of polynomials on modeling accuracy. Case I and Case II, respectively, show the results of parameter estimation with the second-order and fourth-order polynomials:

Case I:

$$\begin{cases} h_{fw} = 3.923 \times 10^{-4}N_e^2 + 0.9841N_e + 703.57, \\ k_1 = 6.08 \times 10^{-3}N_e^2 - 11.894N_e + 21204.1, \\ k_2 = -1.75 \times 10^{-10}N_e^2 + 4.07 \times 10^{-7}N_e + 3.63 \times 10^{-4}, \\ l = -1.528 \times 10^{-7}N_e^2 + 3.4 \times 10^{-4}N_e + 1.08. \end{cases}$$

Case II:

Table 5 presents the R^2 of static parameters for different modeling orders, and it indicates that the higher the modeling order is, the better the fitting accuracy is.

5.2. Nonlinear function identification

There are two nonlinear functions that need to be identified: $g(p_m)$ in differential pressure of steam model (12) and $f(p_{st}, h_{st})$ in superheated steam flow model (13). Procedure of function identification for differential pressure is similar to the regression method in Fig. 3. For differential pressure model, input variable is steam pressure in separator, p_m , and output variable is the differential pressure, $\Delta p = p_m - p_{st}$, where p_{st} is the main steam pressure. Using the variables p_m and p_{st} given in Table 2, the differential pressure $\Delta p = g(p_m)$ is identified as below:

$$\Delta p = g(p_m) = 0.0522p_m - 0.00281, \quad (29)$$

and its R^2 is 0.998.

Referring to Ref. [1], the structure of $f(p_{st}, h_{st})$ in superheated steam flow model (13) is defined as $f(p_{st}, h_{st}) = \frac{ap_{st}}{h_{st}+b}$ with constants a and b . Combining large-scale running data and immune genetic algorithm [1,12], the function can be identified as

$$f(p_{st}, h_{st}) = \frac{145191.5p_{st}}{h_{st} - 199.47}. \quad (30)$$

5.3. Dynamic parameter identification

Dynamic parameters, τ , c_0 and c_3 in the model (16) cannot be identified with steady-state data, but need to be estimated with dynamic data. As presented in (9)–(11), the parameters, c_1 , c_2 , d_1 and d_2 can be calculated by using the basic information in units and IAPWS-IF97 [21], and given as below:

Table 5

R^2 of static parameters in different conditions.

	h_{fw}	k_1	k_2	l
Cubic equation	0.99536	0.706	0.99962	0.882
Case I	0.98998	0.557	0.99740	0.881
Case II	0.99916	0.909	0.99962	0.977

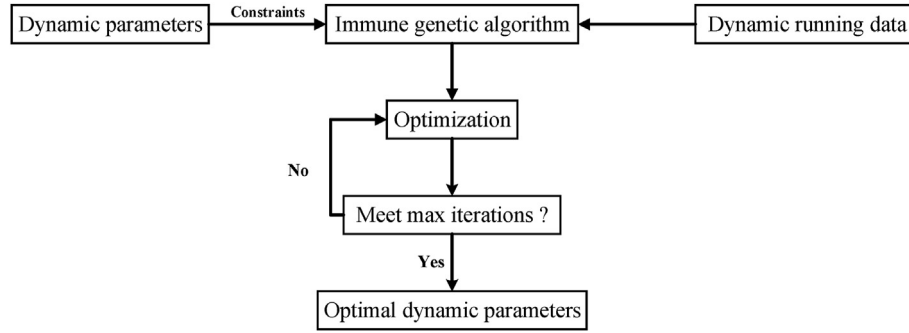


Fig. 4. The procedure of dynamic parameter estimation.

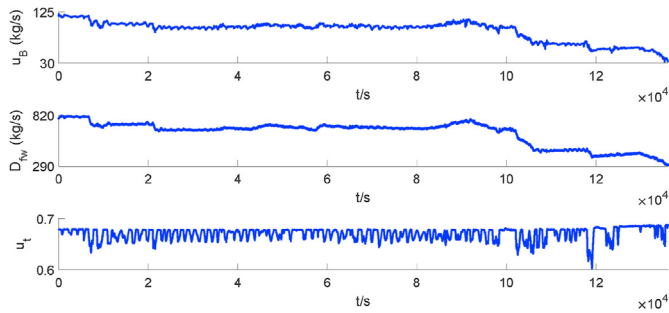


Fig. 5. Input data used for parameter optimization.

parameters can provide a satisfactory dynamic modeling accuracy in a wide load range. In addition, it is interesting and necessary to analyze the model performance based on static parameters in Case I and II, and relevant contents are presented as follows:

Static parameters in Case I are modeled based on a quadratic polynomial, and the parameters, especially k_1 , do not have good fitting accuracy in the comparison shown in Table 5. Therefore, model performance in Case I is not satisfactory as presented in Fig. 6, but the quadratic equation has smooth trends so that calculated outputs do not diverge to infinity after many iterations.

Static parameters in Case II have the highest fitting accuracy as shown in Table 5, but the calculated outputs diverge to infinity in Fig. 7 due to the high-order items in static parameters. In Fig. 7, the function of static parameters in Case II can enlarge the disturbances

$$\begin{aligned}
 c_1 &= -639(-0.005248p_m h_m - 0.003204h_m + 19.7835p_m + 9.7266) \frac{-0.005248h_m + 19.7835}{-0.005248p_m - 0.003204}, \\
 c_2 &= 639(-0.005248p_m h_m - 0.003204h_m + 19.7835p_m + 9.7266), \\
 d_1 &= h_m + \frac{-0.005248p_m h_m - 0.003204h_m + 19.7835p_m + 9.7266}{-0.005248p_m - 0.003204} + \frac{c_j m_j - 1.8113 \cdot 10^{-3} p_m + 0.2012}{639 - 0.005248p_m - 0.003204}, \\
 d_2 &= h_m + \frac{c_j m_j - 1.8113 \cdot 10^{-3} h_m + 11.143}{639 - 0.005248h_m + 19.7835},
 \end{aligned}$$

where $c_j = 0.46 \text{ kJ}/(\text{kg} \cdot ^\circ\text{C})$ and $m_j = 700,000 \text{ kg}$ in this study.

Immune genetic algorithm [1,12] is used to optimize the dynamic parameters, τ , c_0 and c_3 , based on dynamic running data and actual physical constraints, and Fig. 4 presents the procedure of dynamic parameter estimation. After optimization, dynamic parameters are estimated as follows:

$$\tau = 20, c_0 = 380, c_3 = 40.$$

Fig. 5 presents input data used for parameter optimization, and Figs. 6 and 7 show the results of optimization and comparison of model performance based on static parameters in different cases. Besides, Fig. 6 presents the absolute errors between measured values and estimated values.

Table 6 shows the error performance of model with static parameters in different cases. As shown in Figs. 5–7 and Tables 5 and 6, the model with static parameters in the cubical polynomial has the highest dynamic modeling accuracy in a wide load range, and its MAREs of P_{st} , h_m and N_e are 0.974%, 0.839% and 1.676%, respectively. It indicates that the identified static parameters and dynamic

or calculated errors so that model outputs diverge to infinity after many iterations.

The static parameters modeled by cubical polynomials not only have good fitting accuracy, but also have proper functional structure to describe the dynamic process accurately and avoid divergence to infinity; and thus it is the best option to build static parameters in this study.

5.4. Identification of change of throttle loss and energy mismatch functions

In the model (16), there are two functions need to be identified, change of throttle loss and energy mismatch functions, and they can be estimated with practical open-loop running data [11]. Model (16) presents their functional structure, namely, $\Delta Q_1 = \frac{\mu}{N_e + \gamma} (D_{fw} - \alpha r_B)$, and $\Delta Q_{loss} = q(\Delta u_t, p_{st}) = \eta \Delta u_t p_{st}$. In ΔQ_1 , there are two constants, μ and γ , and a coefficient, α , that need to be estimated. The coefficient α can be obtained based on the steady-state data in Table 4 with linear regression method in Fig. 3. The result is given as below:

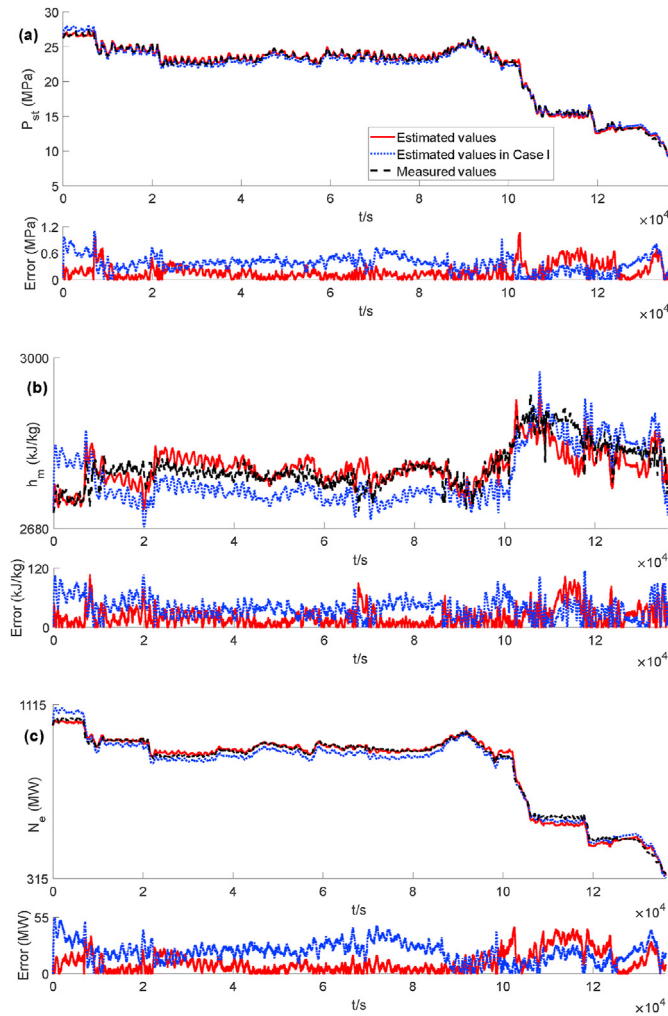


Fig. 6. Results of optimization and model performance with static parameters in Case I. (a) Comparison of the main steam pressure, (b) Comparison of the steam enthalpy in separator, and (c) Comparison of the unit load.

$$\alpha = -2.24 \times 10^{-8} N_e^3 + 5.182 \times 10^{-5} N_e^2 - 3.851 \times 10^{-2} N_e + 16.075. \quad (31)$$

By combining open-loop running data [11] and immune genetic algorithm [1,12], constants μ and γ in ΔQ_1 are identified as $\mu = 334,496.1$ and $\gamma = 171.4$.

For the change of throttle loss, ΔQ_{loss} , the coefficient is identified as $\eta = 8$ based on running data [11] and estimation procedure shown in Fig. 4.

In addition, for safety and economic consideration, nonlinear constraints of model inputs are presented as follows:

$$u_B \in [30 \text{ kg/s}, 120 \text{ kg/s}], \quad D_{fw} \in [-50 + \alpha r_B \text{ kg/s}, 50 + \alpha r_B \text{ kg/s}], \quad u_t \in [0.5, 1]. \\ |\Delta u_B| \leq 10 \text{ kg/s}, \quad |\Delta D_{fw}| \leq 40 \text{ kg/s}, \quad |\Delta u_t| \leq 1\%.$$

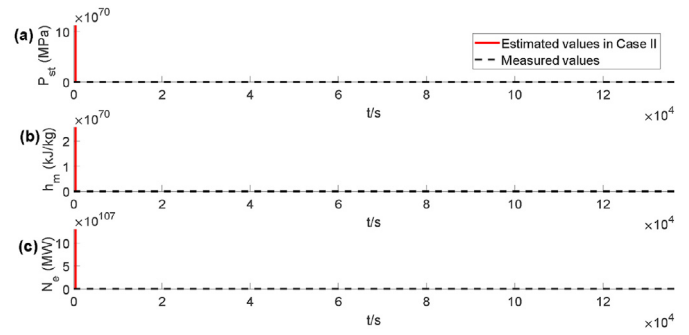


Fig. 7. Model performance with static parameters in Case II. (a) Dynamic responses of the main steam pressure, (b) Dynamic responses of the steam enthalpy in separator, and (c) Dynamic responses of the unit load.

5.5. Discussion

Compared with previous work [1,5,11–13], static parameters in this work are functions of unit load, and these parameters can show their physical information based on current output variables. Furthermore, the information can be used to calculate model inputs in steady-state conditions, and the inputs can be viewed as feed-forward to improve control performance. For dynamic parameters, previous researches regard them as constant variables that need to be identified by combining optimization algorithm and running data. In this case, their physical meanings are ignored, which can deteriorate model performance. Moreover, the method can increase the computational burden of dynamic parameter optimization. In this work, dynamic parameter in boiler systems can be derived from physical information and IAPWS-IF97, which reduces computation burden and enriches the physical significance of model established.

6. Validation

To test model performance, this section presented steady-state and dynamic validations to compare the modeling accuracies, and open-loop simulation and validation were conducted to obtain model performance in abnormal conditions. In addition, input set points are generated from given output set points, and nonlinear model predictive control (NMPC) was used to design a control system and show its control performance.

5.1. Steady-state validation

Validation was performed to obtain the steady-state accuracy of the model established, and Table 7 presents the estimated values and relative errors of the model outputs.

As shown in Table 7, the maximum and minimum relative errors of the main steam pressure are 4.78% and 0.05%, respectively. Similarly, those of steam enthalpy in separator are respectively

Table 6

Error performance of model with static parameters in a wide load range.

Error	P_{st} (MPa)			h_m (kJ/kg)			N_e (MW)		
	Cubic equation	Case I	Case II	Cubic equation	Case I	Case II	Cubic equation	Case I	Case II
MAE	0.184	0.357	NaN	23.55	37.60	NaN	11.86	20.91	NaN
MARE(%)	0.974	1.681	NaN	0.839	1.346	NaN	1.676	2.604	NaN
RMSE	0.245	0.400	NaN	31.6	43.14	NaN	16.09	23.20	NaN

Table 7

Estimated values and relative errors of model outputs at different loads.

Estimated values			Relative errors (%)		
p_{st} /MPa	h_m /(kJ.kg ⁻¹)	N_e /MW	δp_{st}	δh_m	δN_e
9.371	2754.3	330.01	0.98	0.51	4.07
11.037	2811.1	402.36	2.92	0.08	3.22
12.388	2855.1	462.60	4.78	1.72	6.64
13.268	2811.3	488.96	0.67	0.58	1.85
15.102	2830.8	565.52	3.12	1.73	5.89
22.992	2811.7	892.87	0.86	0.74	1.69
24.821	2770.9	951.99	0.05	0.70	0.05
26.706	2732.6	1034.90	0.66	0.22	1.18

1.73% and 0.08%, and those of unit load are 6.64% and 0.05%, respectively. Due to fitting errors, the model established does not have satisfactory accuracies at 330.01 MW, 462.6 MW and 565.52 MW loads, but the model has generally acceptable steady-state accuracies to describe the dynamic operational process in a wide load range.

5.2. Dynamic validation

Two groups of running data of the 1000 MW USC-OTBT unit were used to validate dynamic modeling accuracy in a wide load range, and the discrete sampling time is 1s. The two groups of running data have different trends shown as below:

Case A: Unit load decreases from 1008.1 MW to 353.02 MW.

Case B: Unit load increases from 341.9 MW to 1030.3 MW.

Coal quality and combustion conditions vary with time in normal operation, and thus parameters in the model also change

with time. As presented in Section 4, the static parameters k_1 and l defined in (21) and (22) do not have good fitting accuracy, and thus this study considers the influence of change of time on these parameters. In addition, two energy functions presented in Section 4.4 are shown to improve dynamic modeling accuracy in comparison with the previous modeling methods [1,5,11–13].

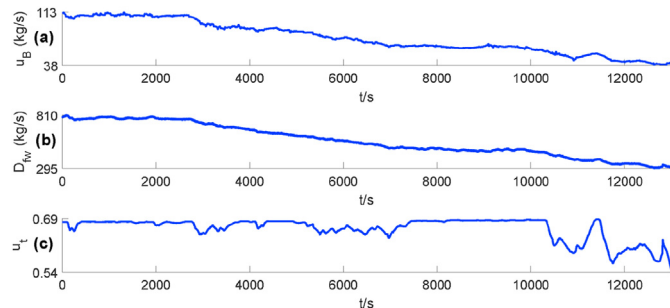
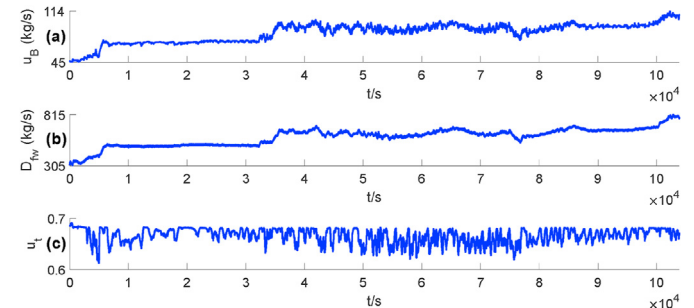
Parameters k_1 and l for Case A and Case B can be identified based on the linear regression method in Fig. 3, and the results are presented below:

Case A:

$$\begin{cases} k_1 = -5.55 \times 10^{-5} N_e^3 + 0.12 N_e^2 - 84.87 N_e + 35634.5, \\ l = 2.888 \times 10^{-10} N_e^3 - 4.072 \times 10^{-7} N_e^2 + 1.8306 \times 10^{-4} N_e + 1.21. \end{cases}$$

Case B:

$$\begin{cases} k_1 = 1.785 \times 10^{-5} N_e^3 - 3.37643 \times 10^{-2} N_e^2 + 16.22 N_e + 15458.14, \\ l = -5.9834 \times 10^{-10} N_e^3 + 9.949 \times 10^{-7} N_e^2 - 3.9853 \times 10^{-4} N_e + 1.2651. \end{cases}$$


Fig. 8. Input data used for dynamic validation in Case A. (a) Coal flow command, (b) Total feedwater flow, (c) Throttle valve opening.

Fig. 9. Input data used for dynamic validation in Case B. (a) Coal flow command, (b) Total feedwater flow, (c) Throttle valve opening.

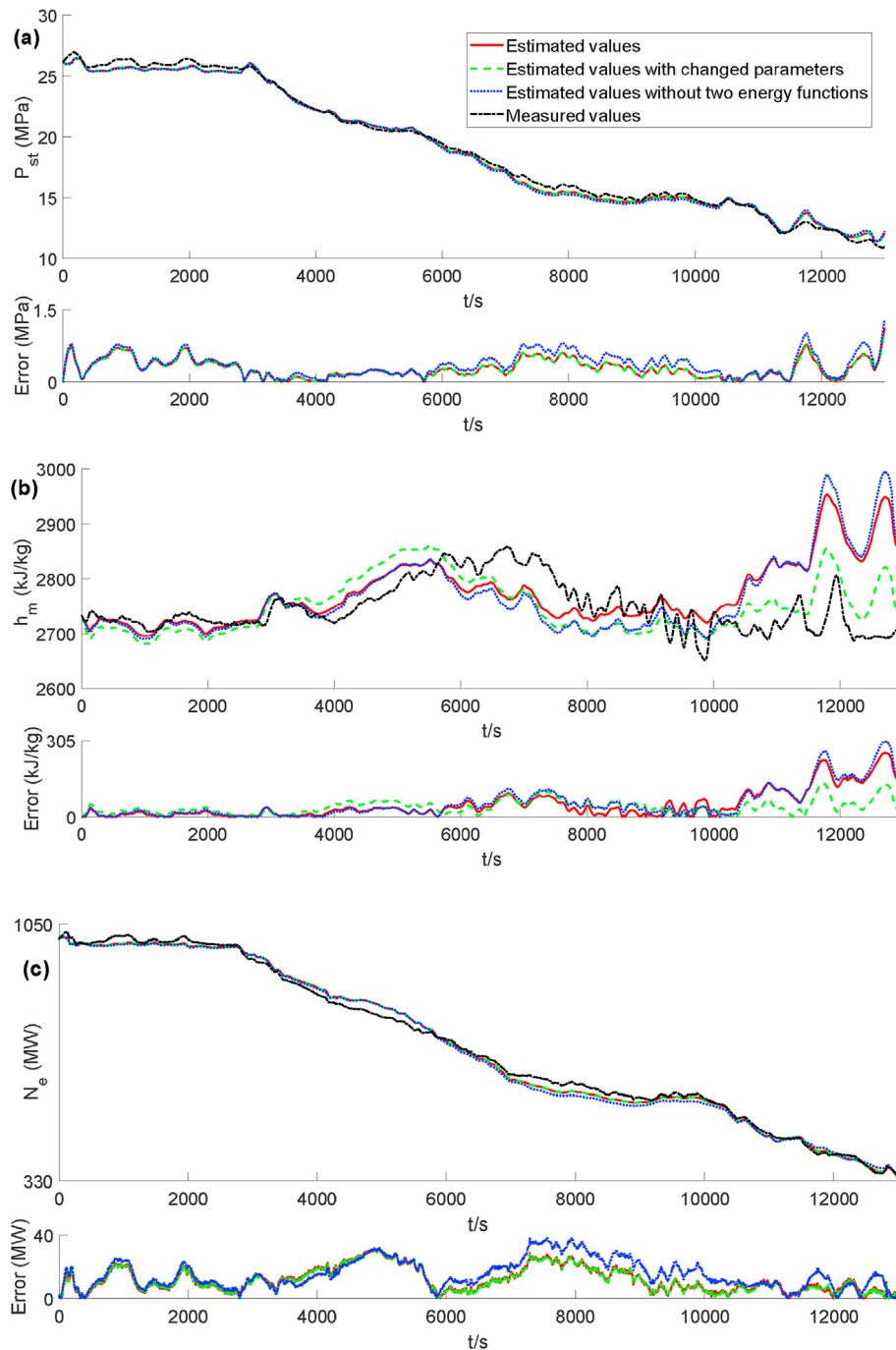


Fig. 10. Estimated and measured values of outputs in Case A. (a) Dynamic responses of main steam pressure, (b) Dynamic responses of steam enthalpy in separator, and (c) Dynamic responses of unit load.

Note that parameters in Case A and B are quite different from those estimated in (26) and (28) in Section 4. Figs. 8 and 9 present the input data used for dynamic validation in Case A and Case B, respectively, and Figs. 10 and 11 show the dynamic model performances with two energy functions and changed parameters, k_1 and l , in Case A and Case B. Besides, error performances of model outputs are shown in Table 8 and Figs. 10 and 11.

As shown in Figs. 8–11 and Table 8, the estimated values of model outputs are in strong agreement with the measured values and have similar dynamic trends with measured values. In addition,

MAREs of model outputs are generally less than 1.8% and 2.85% in Case A and Case B, respectively, and thus the model established shows the acceptable dynamic accuracy in a wide load range from low load to full load. It also indicates that parameters and functions identified in this work can capture the essential operational characteristics of USC-OTBT units in a wide load range.

In practice, operational conditions change with time, and thus this work considers the influence of change of time on model performance. Static parameters, k_1 and l , are different in Case A and Case B, and the model performances are presented in Figs. 10 and 11 and Table 8. By comparison, k_1 and l in Case A and Case B can

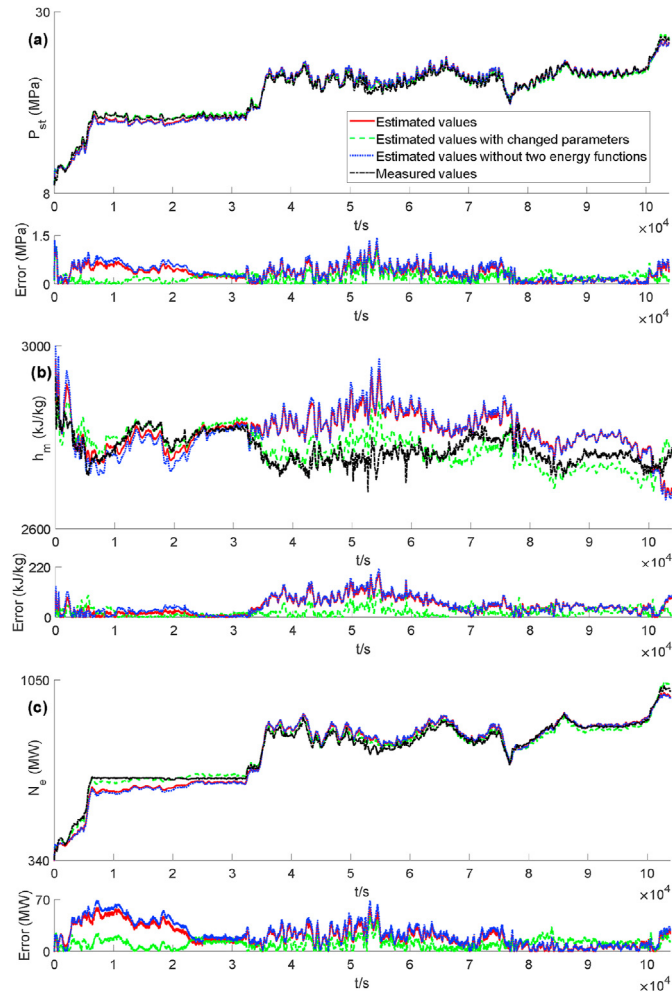


Fig. 11. Estimated and measured values of outputs in Case B. (a) Dynamic responses of main steam pressure, (b) Dynamic responses of steam enthalpy in separator, and (c) Dynamic responses of unit load.

describe the operational characteristics in steady-state conditions accurately, and thus the model outputs with changed parameters have the satisfactory dynamic accuracy in a wide load range. MAREs of model outputs are generally less than 1.7% and 1.4% in Case A and Case B. It also indicates that the change of time has an important influence on k_1 and l , and other parameters and functions are relatively insensitive to the change of time.

From Figs. 10 and 11 and Table 8, the estimated values of the model have the higher dynamic accuracy than those without two

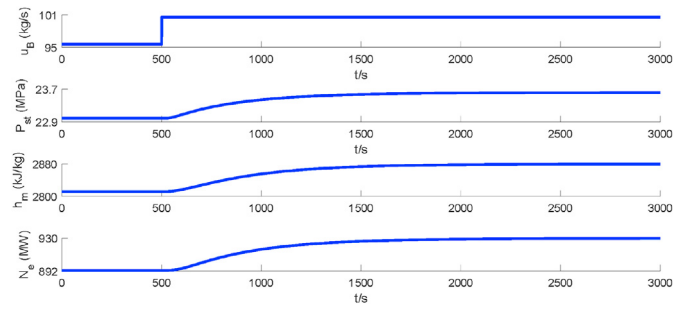


Fig. 12. Dynamic responses of model outputs to step increase in coal flow command.

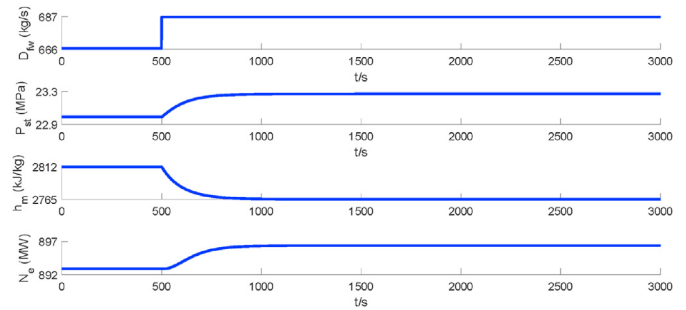


Fig. 13. Dynamic responses of model outputs to step increase in total feedwater flow.

energy functions [1,5,11–13] in Case A and Case B, and it indicates that change of throttle loss and energy mismatch functions can describe the operational process of USC-OTBT units accurately and improve dynamic modeling accuracy.

Due to inevitable modeling errors, it is meaningful to find more mechanism knowledge or proper model structure in order to further reduce modeling errors and improve modeling accuracy in future. Moreover, the concept of change of throttle loss and energy mismatch functions can also be applied into OTBT and other units to improve model performance.

5.3. Open-loop simulation and validation

Open-loop simulation and validation are conducted to show the correctness of the model established, and the simulation results are presented in Figs. 12–14.

As presented in Fig. 12, when coal flow command increases quickly, main steam pressure, steam enthalpy in separator and unit load gradually increase to the higher level due to more heat energy released from coal combustion in furnace.

When total feedwater flow increases rapidly, Fig. 13 shows the

Table 8

Mean absolute relative errors (MARE) and root mean square errors (RMSE) of outputs.

Case	Method		p_{st} (MPa)	h_m (kJ/kg)	N_e (MW)
Case A	Estimated values	MARE (%)	1.55	1.79	1.71
		RMSE	0.333	74.42	14.07
	Estimated values with changed parameters	MARE (%)	1.55	1.41	1.67
		RMSE	0.333	47.20	13.89
	Estimated values without two energy functions	MARE (%)	1.99	1.95	2.39
		RMSE	0.413	80.08	18.26
Case B	Estimated values	MARE (%)	1.74	1.71	2.84
		RMSE	0.40	59.71	24.72
	Estimated values with changed parameters	MARE (%)	1.00	0.84	1.38
		RMSE	0.24	29.72	12.35
	Estimated values without two energy functions	MARE (%)	2.02	1.86	3.38
		RMSE	0.46	64.19	29.35

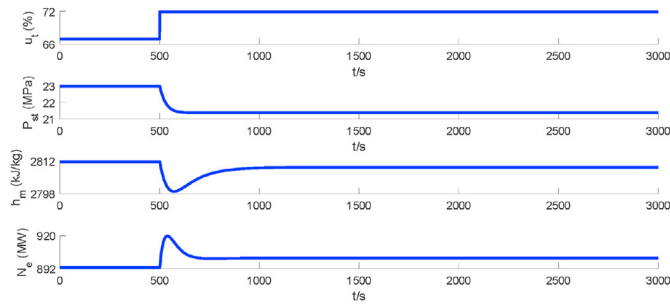


Fig. 14. Dynamic responses of model outputs to step increase in throttle valve opening.

dynamic responses of model outputs. As the pressure of feedwater flow increases, steam in boiler is squeezed and then main steam pressure rises to a higher level quickly. Similarly, as more steam is sent to turbine, unit load increases, and then gradually decreases to a level higher compared with the original level due to more heat energy absorbed by steam. Besides, for more feedwater in boiler, steam enthalpy in separator gradually decreases to a lower level.

As presented in Fig. 14, when throttle valve opening increases suddenly, main steam pressure decreases to a lower level due to the enlarged flow area. Similarly, steam enthalpy in separator decreases quickly due to the decline of steam pressure, and finally restores to the original level due to unchanged coal flow and feedwater flow. As throttle valve opening increases to release heat energy stored in boiler, more steam is injected into turbine and increase the unit load. After the release, unit load gradually decreases to the level higher than the original value due to the enlarged throttle valve.

The dynamic responses of model outputs are in accord with operational experience, and thus the model established can describe the open-loop operational characteristics of USC-OTBT units well qualitatively.

Based on the open-loop running data of USC-OTBT units [11], open-loop simulations were performed to verify the open-loop performance and analyze the dynamic characteristics. In practice, open-loop experiments are performed at steady-state operational condition, and there is a one-to-one match between model inputs, u_B , D_{fw} , u_t , and model outputs, p_{st} , h_m , N_e , in steady-state condition. For different USC-OTBT units, model outputs of the units are not very consistent for the same model inputs, and thus open-loop running data [11] in an incremental form are utilized to validate the open-loop performance of the model. In addition, models in previous studies [1,5,11–13] ignore the effect of change of throttle loss and energy mismatch functions, and thus open-loop responses of model without two energy functions are presented and compared to demonstrate the effectiveness of the proposed model. The results are given in Figs. 15–17, and mean absolute errors (MAEs) of outputs are listed in Table 9.

In Fig. 15, when coal flow command decreases by roughly 3 kg/s quickly, real main steam pressure, steam enthalpy in separator and unit load gradually decline by nearly 0.5 MPa, 30 kJ/kg and 25 MW, respectively. As coal flow in furnace descends and feedwater flow keeps unchanged, the temperature of working fluid is lower than the temperature in the nominal operation, and thus extra heat energy described by energy mismatch function in Q_1 , is absorbed by working fluid due to the larger temperature deviation between working fluid and metal tube. As shown in Fig. 15, the responses of the model established are closer to real responses as compared with the responses of model without the two energy functions.

Similarly, with the increase of feedwater flow in Fig. 16, the temperature of working fluid is lower than the temperature in the nominal operation, and then working fluid absorbs more heat

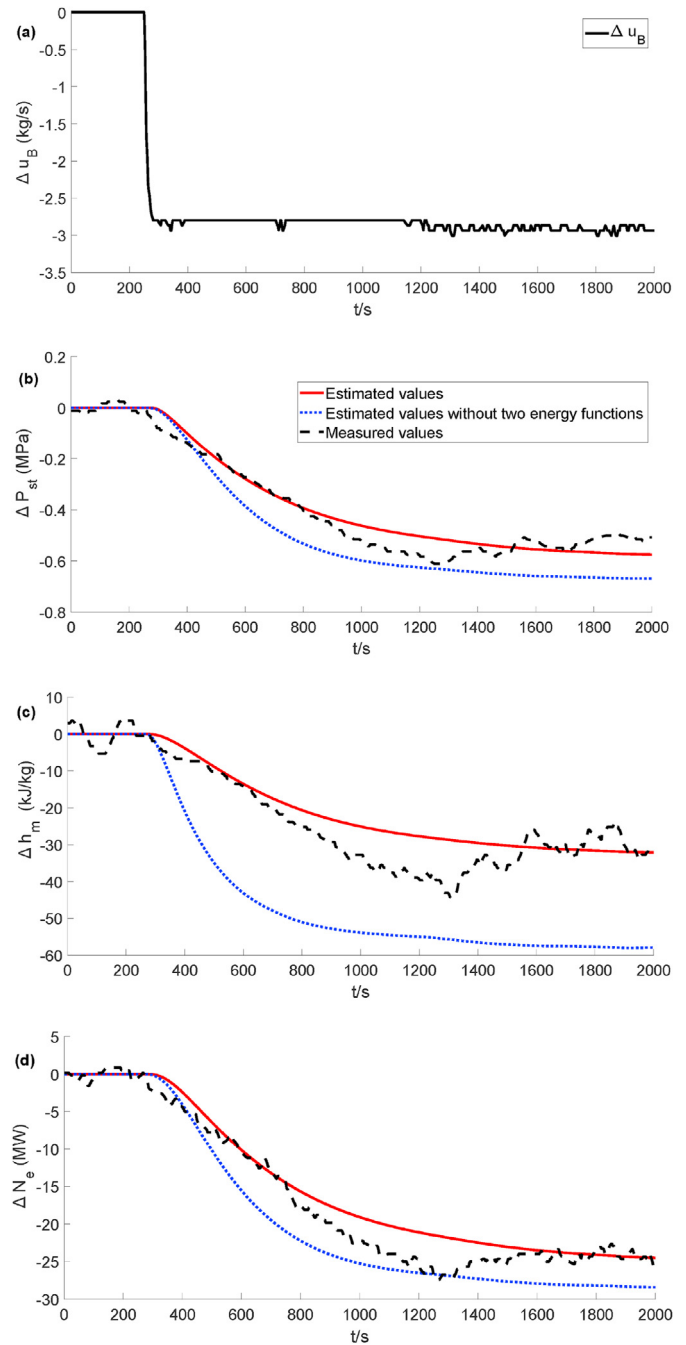


Fig. 15. Open-loop responses of model outputs to coal flow command under 788 MW load condition. (a) Responses of the coal flow command, (b) Responses of the main steam pressure, (c) Responses of the steam enthalpy in separator, and (d) Responses of the unit load.

energy provided by the energy mismatch function in Q_1 . Consequently, real main steam pressure and unit load gradually rise by approximately 0.25 MPa and 3 MW, respectively, and steam enthalpy in separator decreases by roughly 30 kJ/kg. Due to extra heat energy, the open-loop responses of the model are closer to real responses as compared with the responses of the model without the energy mismatch function.

As the throttle valve opening decreases by approximately 0.065 shown in Fig. 17, the main steam pressure gradually increases by 1.2 MPa due to decreased flow area. Subsequently, with the increase of steam pressure, steam enthalpy in separator rises, and then the

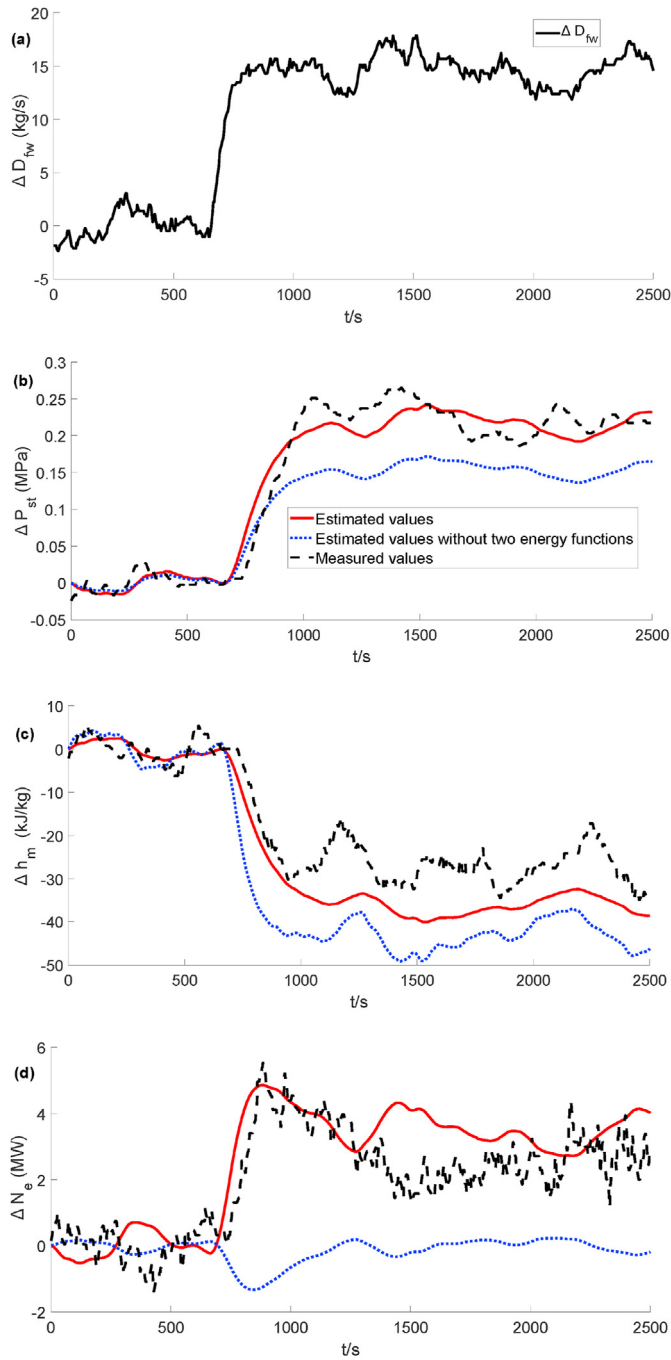


Fig. 16. Open-loop responses of model outputs to feedwater flow under 800.7 MW load condition. (a) Responses of the feedwater flow, (b) Responses of the main steam pressure, (c) Responses of the steam enthalpy in separator, and (d) Responses of the unit load.

enthalpy is gradually restored to the nearly original value due to the unchanged ratio of water to coal. Importantly, as throttle valve opening decreases and then throttle loss increases, the unit load decreases quickly and gradually rises to the new level lower than the original. As the function, ΔQ_{loss} , describes the change of throttle loss, the responses of model established are in accord with practical responses as compared with the previous model performance without considering the throttle loss.

To be specific, Table 9 presents the MAEs of two modeling methods to compare their modeling accuracies in abnormal

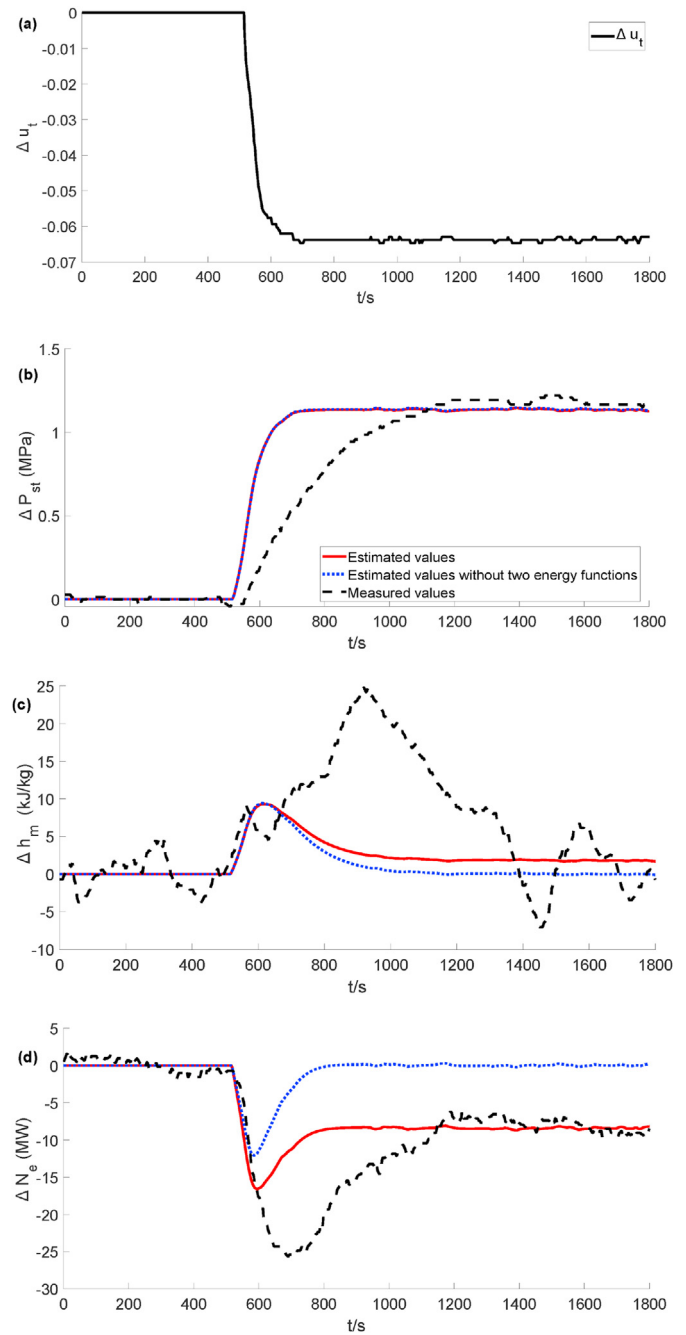


Fig. 17. Open-loop responses of model outputs to throttle valve opening under 612.7 MW load condition. (a) Responses of the throttle valve opening, (b) Responses of the main steam pressure, (c) Responses of the steam enthalpy in separator, and (d) Responses of the unit load.

operational conditions. For the responses to coal flow command and feedwater flow, MAEs of three model outputs in this study are reduced by at least 50% in comparison with the responses of model without energy mismatch function. As for the responses to throttle valve opening, throttle loss only affects unit load (N_e) according to its physical definition, and thus the MAE of unit load is decreased by more than 50% as shown in Table 9. However, MAEs of model outputs, p_{st} and h_m , almost remain constant in both responses.

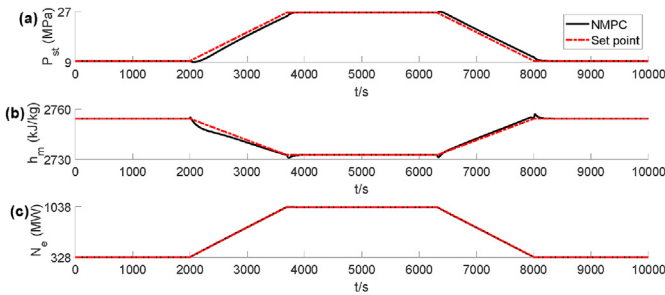
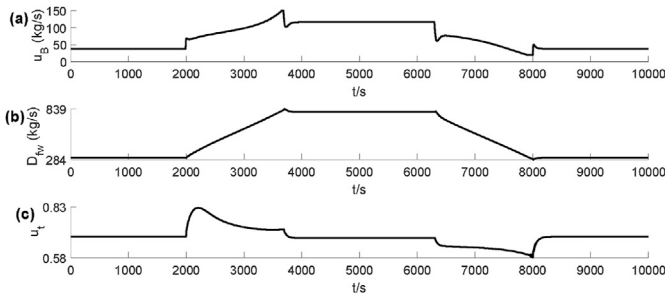
As mentioned above, the two energy functions, change of throttle loss and energy mismatch functions, can improve the model structure and open-loop modeling accuracy, and thus the

Table 9
Mean absolute errors of model outputs.

Method	Input	$p_{st}/(\text{MPa})$	$h_m/(\text{kJ}\cdot\text{kg}^{-1})$	$N_e/(\text{MW})$
Improved responses	Coal flow command	0.0381	4.6613	1.4855
	Feedwater flow	0.0234	6.5751	0.8936
	Throttle valve opening	0.1236	5.8191	2.9464
Previous responses	Coal flow command	0.0892	19.584	2.796
	Feedwater flow	0.0476	12.496	2.160
	Throttle valve opening	0.1240	6.3841	7.9466

Table 10
Parameters in NMPC system.

N	N_u	μ_1	μ_2	μ_3	λ_1	λ_2	λ_3
60	2	8000	100	100	1	1	1

**Fig. 18.** Output responses of NMPC system based on the model established: (a) Response of main steam pressure, (b) Response of steam enthalpy in separator, and (c) Response of unit load.**Fig. 19.** Input responses of NMPC system based on the model established: (a) Response of coal flow command, (b) Response of feedwater flow, and (c) Response of throttle valve opening.

model established can describe the operational characteristics of USC-OTBT units in abnormal conditions well quantitatively.

5.4. Input set points in steady-state conditions

Compared with previous studies [1,5,11–13], this model can provide more operational information in steady-state conditions so that input set points can be generated from given output set points, which then can be viewed as feedforward signals to improve control system performance.

Specifically, input set points can be determined and presented as below:

- (1) Determine output set points, namely, p_{str} , h_{mr} and N_{er} .
- (2) As static parameters are nonlinear functions of unit load, the static parameters set point, namely, k_{1r} , k_{2r} , l_r , h_{fwr} and α_r , can be calculated based on unit load set point, and main steam enthalpy set point can also be obtained, namely, $h_{str} = l_r h_{mr}$.
- (3) Based on (18) and (20), feedwater flow set point can be calculated as below:

$$u_{2r} = \frac{N_{er}}{k_{2r}(h_{str} - h_{fwr})}. \quad (34)$$

- (4) Similarly, by combining (13) and (18), throttle valve opening set point can be calculated as below:

$$u_{3r} = \frac{h_{str} + b}{a p_{str}} \frac{N_{er}}{k_{2r}(h_{str} - h_{fwr})}. \quad (35)$$

- (5) In steady-state conditions, change of throttle loss doesn't exist due to no change of throttle valve opening, but energy mismatch function is considered to obtain the accurate coal flow command set point. By combining (4), (8) and (20), the set point of coal flow command is obtained as below:

$$u_{1r} = \frac{\frac{N_{er}}{k_{2r}} - \frac{\mu}{N_{er} + \gamma} \frac{N_{er}}{k_{2r}(h_{str} - h_{fwr})}}{k_{1r} - \frac{\mu}{N_{er} + \gamma} \alpha_r}. \quad (36)$$

5.5. A case for control system design based on NMPC

To obtain the control performance, nonlinear model predictive control (NMPC) [22] is adopted in this study to design control system based on the model established.

In NMPC system, the set point of unit load increases from 330 MW to 1035 MW, and then decreases from 1035 MW to 330 MW. Simulation time is 10,000s, and sample time is 1s. Subsequently, load rate is 25 MW/min, and Table 10 shows the parameters in the NMPC system.

In Table 10, N and N_u are predictive and control horizons, respectively, and weights μ_i and λ_i ($i = 1, 2, 3$) are applied respectively to predictive errors of controlled variables and increments of manipulated variables. Besides, increments of manipulated variables in an explicit form are adopted to reduce the computational cost, and rate constraints are presented as below:

$$-10 \leq \Delta u_1 \leq 10, \quad -40 \leq \Delta u_2 \leq 40, \quad -0.01 \leq \Delta u_3 \leq 0.01. \quad (37)$$

The cost function of NMPC is defined as below:

$$J = \sum_{p=1}^N \|y_r(k+p|k) - y(k+p|k)\|_{M_p}^2 + \sum_{p=0}^{N_u-1} \|\Delta u(k+p|k)\|_{\Lambda_p}^2, \quad (38)$$

where k is current time, and $M_p = \text{diag}(\mu_1, \mu_2, \mu_3)$ and $\Lambda_p = \text{diag}(\lambda_1, \lambda_2, \lambda_3)$ are weight matrices.

In (38), $y_r(k+p|k)$ and $y(k+p|k)$ are set points and predicted values of controlled variables on the prediction horizon at the current time k , respectively, where controlled variables are $y(t) = [P_{st}(t), h_m(t), N_e(t)]$, and $\Delta u(k+p|k)$ is the increment of manipulated variables on the control horizon at current time k , where manipulated variables are $u(t) = [u_B(t-\tau), D_{fw}(t), u_t(t)]$.

Figs. 18 and 19 present the simulation results, and, to obtain the control performance, the RMSE is used as defined below:

$$RMSE_i = \sqrt{\frac{1}{N_{sim}} \sum_{k=1}^{N_{sim}} (y_i^{sp}(k) - y_i(k))^2}, \quad i = 1, 2, 3, \quad (39)$$

where $y_i^{sp}(k)$ denotes the set points of model outputs and $y_i(k)$ are the calculated values of output variables during simulations, N_{sim} is the simulation horizon.

As shown in Figs. 18 and 19, calculated values in NMPC system can track the output set points quickly, and $RMSE_1$, $RMSE_2$, and $RMSE_3$ are 0.7431 MPa, 0.916 kJ/kg and 1.31 MW, respectively. It indicates that NMPC system has satisfactory tracking performance based on the model established. From input responses, coal flow command increases rapidly to provide more heat energy in furnace when load demand rises, and feedwater flow matches coal flow and then increases to maintain safe and economical operation. When the deviation of unit load is reduced, the upward trend of coal flow command gradually decreases and keeps unchanged. As for the response of throttle valve opening, the opening enlarges quickly to release heat energy stored in boiler system in order to track the rising load demand, and then the opening gradually decreases to maintain main steam pressure stable. Accordingly, the input responses are in accord with operational experience, and it demonstrates that the model can be used for control system design to test various control algorithms.

7. Conclusion

This study adopts the first-principle modeling method to develop a dynamic nonlinear mathematical model for a wide-load range operation of ultra-supercritical (USC) once-through boiler-turbine (OTBT) units. Parameters and functions in the model are identified by combining dynamic running data, open-loop running data [11] and an optimization algorithm [12]. Subsequently, validation and simulation are conducted to demonstrate the model performance. Specific conclusions in this study are drawn as below:

- (1) Nonlinear characteristics of USC-OTBT units in low load are analyzed, and the model established can provide acceptable dynamic accuracies of the units in a wide load range from low load to full load.
- (2) Change of throttle loss and energy mismatch functions are built to improve model structure and dynamic accuracy, and validation results illustrate that the model can describe the dynamic process of USC-OTBT units in normal and abnormal conditions accurately.
- (3) The model can describe more steady-state operational characteristics so that input set points can be generated from given output set points to improve control performance.

- (4) The NMPC is designed based on the model established, and the results illustrate that control responses are in accord with operational experience and that the model can be used for control system design.

Based on the above work, there are two open problems to solve in future: 1) It is necessary to analyze disturbances in practical operation and then further improve modeling accuracy of USC-OTBT units, 2) Advanced control algorithms can be used to test the model and further improve the CCS performance.

Credit author statement

He Fan: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing – original draft, Visualization, Supervision, Project administration; Zhi-gang Su: Writing – review & editing, Funding acquisition; Pei-hong Wang: Writing – review & editing; Kwang Y. Lee: Writing – review & editing.

Declaration of competing interest

All authors declare that no conflict of interest exists.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.energy.2021.120425>.

Data Availability

In this work, running data for USC-OTBT units in a wide load range are provided in Mendeley data and IEEE Dataport, and their addresses are presented as below:

- [1] Fan, He (2021), "Running data for USC-OTBT units in a wide load range", Mendeley Data, V1, doi: <https://doi.org/10.17632/3gbyckytyj.1>.
- [2] He Fan, March 7, 2021, "Running data for USC-OTBT units in a wide load range", IEEE Dataport, doi: <https://dx.doi.org/10.21227/ma4d-9j43>.

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